

JEE MAIN REVISION MASTER: MOVING CHARGES AND MAGNETISM

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 48 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Biot-Savart Law & Magnetic Field Calculations	41 PYQs	Magnetic Fields
Moderate	Lorentz Force, Galvanometer & Conversion to Ammeter / Voltmeter	7 PYQs	Magnetic Forces & Devices

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Biot-Savart Law & Magnetic Field Calculations	$B_{\text{center}} = \frac{\mu_0 N I}{2 R}, \quad B_{\text{wire}} = \frac{\mu_0 I}{2\pi d}, \quad B_{\text{solenoid}} = \mu_0 n I \quad (n = N/L)$
Lorentz Force, Galvanometer & Conversion to Ammeter / Voltmeter	$r = \frac{mv}{qB}, \quad \text{Shunt: } S = \frac{I_g G}{I - I_g}, \quad \text{Multiplier: } R = \frac{V}{I_g} - G$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Biot-Savart Law & Magnetic Field Calculations (Very High Yield — 41 PYQs)

Core Principle: Biot-Savart law: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^2}$. Magnetic field at center of circular loop $B = (\mu_0 N I / 2R)$. On the axis of circular coil: $B = \frac{\mu_0 N I R^2}{R^2 + x^2}^{3/2}$. Infinitely long straight wire: $B = (\mu_0 I / 2\pi d)$. Inside a long solenoid: $B = \mu_0 n I$.

■ **Caution / JEE Pitfall:** At the ends of a semi-infinite solenoid, magnetic field drops to exactly HALF: $B_{\text{end}} = (1/2)\mu_0 n I$.

■ **Speed Trick:** If a circular wire of radius R is bent into n turns of radius R/n , the magnetic field at the center increases by n^2 times!

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this cha..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Two identical circular current loops carrying equal currents are placed with their axes inclined at 45° to each other as shown in t..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Biot-Savart law indicates that an electron moving with a velocity $\vec{v}$ produces a magnetic field $\vec{B}$ around it such that..." [Ans: N/A]

3.2 Lorentz Force, Galvanometer & Conversion to Ammeter / Voltmeter (Moderate Yield — 7 PYQs)

Core Principle: Lorentz force $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. Circular radius $r = (mv / qB)$, period $T = (2\pi m / qB)$ (independent of speed v). Force between parallel wires: $(F / L) = (\mu_0 I_1 I_2 / 2\pi d)$ (parallel currents attract).

Galvanometer conversion: Ammeter implies small shunt $S = (I_g G / I - I_g)$ in parallel; Voltmeter implies high resistance $R = (V / I_g) - G$ in series.

■ **Caution / JEE Pitfall:** Ammeter is connected in SERIES and must have very low resistance (ideal = 0). Voltmeter is connected in PARALLEL and must have very high resistance (ideal = ∞).

■ **Speed Trick:** To increase the range of an ammeter by n times ($I = n I_g$), the required shunt is simply $S = (G / n - 1)$.

Representative Source PYQs:

- **[2025 Shift] Q#1 (MCQ):** "Two thin long parallel wires separated by a distance 'r' from each other in vacuum carry a current of I ampere in opposite directions. T..." **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "A charged particle is subjected to acceleration in a cyclotron as shown. The charged particle undergoes increase in its speed..." **[Ans: N/A]**
- **[2021 Shift] Q#1 (MCQ):** "A strong magnetic field is applied on a stationary electron. Then, the electron..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this charge. Then

- (A) observes both electric and magnetic fields
- (B) observes both electric and magnetic fields
- (C) observes only magnetic field
- (D) observes only electric field

Correct Answer: None

Solution Analysis: The correct option is: Option B: B observes both electric and magnetic fields. The point charge is placed in the moving train. For passenger A sitting inside the train, the charge is at rest. A stationary charge produces only an electric field around it. So A observes only electric field, not magnetic field. For person B standing on the ground, the train is moving. So the charge is also moving with the train. A moving charge is equivalent to an electric current, and current produces a magnetic field. Since the charge itself also has electric field, B observes both electric and magnetic

Benchmark Problem 2: KCET 2023 (Q#3)

A charged particle is subjected to acceleration in a cyclotron as shown. The charged particle undergoes increase in its speed

- (A) Only in the gap between $D_{(1)}$ and $D_{(2)}$
- (B) Only inside $D_{(2)}$
- (C) Inside $D_{(1)}$, $D_{(2)}$ and the gaps
- (D) Only inside $D_{(1)}$

Correct Answer: None

Solution Analysis: A charged particle experiences two different forces in a cyclotron as follows. (i) Inside the Dees It is accelerated due to the magnetic field inside the dees. (ii) In the Gaps It is accelerated due to the electric field.